Logan Diep

logan.diep1@gmail.com | linkedin.com/in/logan-diep | Portfolio Website

Education

Stony Brook University — **Aug. 2023 – May 2027**
Bachelor's of Engineering in Computer Engineering | GPA: 3.6 — *Stony Brook, New York*

Relevant Coursework

- Computer Architecture
- Digital Design Using VHDL
- Embedded MCU Systems Design
- Electrical Circuit Analysis
- Real-Time Operating Systems
- Data Structures & Algorithms

Technical Skills

Languages: SystemVerilog, VHDL, C++, C, Embedded C, AVR Assembly, Python, MATLAB
Developer Tools: Altium Designer, Xilinx Vivado, Aldec Active-HDL, Git, LTspice, VS Code, OrCAD Capture CIS, KiCad, Arduino IDE, Microchip Studio
Lab Tools: Digital Multimeter, Oscilloscope, Bench PSU, Signal Generator, Spectrum Analyzer, Logic Analyzer

Experience

BAE Systems, Inc. — **Incoming Summer 2026**
Electrical Engineering Intern — *Wayne, NJ*

- Electronic Systems Hardware Engineering Department

SRCTec, LLC — **May 2025 – Aug. 2025**
Test Engineering Intern — *Syracuse, NY*

- Built custom automated test equipment (ATE) incorporating interface test adapter (ITA) modules to automate board-level diagnostics, executing repeatable voltage-rail checks, loopback, and I2C/SPI communication tests.
- Standardized LCMR radar test CSVs into JSON and integrated with WATS (Virinco) to enable per-board/test analytics, revealing recurring issues in loopback, voltage, and SPI/I2C/UART tests and prompting targeted design updates.
- Created interconnection diagrams in Visio for test stations, documenting PC ports, bench-PSU rails, wave-generator channels, DMM measurement points, and logic-analyzer probes, enabling repeatable board-specific station builds.

Stony Brook Robotics Team — **May 2025 – Present**
Electrical Team Lead — *Stony Brook, NY*

- Leading electrical for MATE ROV, owning vehicle power and control architecture, running weekly design reviews, and coordinating integration with mechanical/software.
- Designing a 6-layer ROV motherboard around an STM32F407 and NVIDIA Jetson, integrating 12V and 5V rail sensing, dual 48V-to-12V power-card interfaces, and 8 ESC thruster channels through modular gold-finger card-slot connectors for scalable power distribution, easier fault isolation, and quick replacement of failed subsystems during debugging.
- Creating the ROV's Systems Integration Diagram (SID) to document power, signals, and interfaces for software/mechanical coordination during bring-up.

DoIT Academic Technology Services (SBU) — **Jan. 2024 – Present**
Lead Student Technologist — *Stony Brook, NY*

- Resolving 100+ TeamDynamix tickets per week for LMS issues (Brightspace, Respondus, Zoom, Echo360).
- Organizing and facilitating ECAD/MCAD workshops, instructing over 60 students in foundational electronics design.

Projects

Single-Cycle RISC-V CPU on FPGA | *SystemVerilog, Xilinx Vivado, Testbenches* — **Aug. 2025 – Present**

- Implementing a single-cycle RV32I CPU in SystemVerilog, designing the datapath, register file, ALU, and control modules with separate instruction/data memories for deployment on a Basys 3 FPGA Development Board.
- Producing SystemVerilog testbenches to validate decoding, datapath, and ALU control through simulation on Vivado.
- Verifying hardware execution on Basys 3 FPGA by compiling RISC-V programs into hex files, loading them into instruction memory, and confirming results via LEDs and seven-segment displays.

Microcontroller-Based Roulette Wheel | *Altium Designer, Power Regulation, C++* — **May 2025 – Jun. 2025**

- Developed an "LED Roulette" PCB featuring an ATmega32U4 microcontroller with native USB programmability.
- Routed a 90 Ω differential pair with 22 Ω series resistors for USB D+/D- signals to meet full-speed USB compliance.
- Added TVS diodes to safeguard USB data lines from ESD and implemented a linear voltage regulator with bulk and high-frequency decoupling to stabilize the USB 5 V rail.

EPL Standings Prediction & Analysis | *Python, Web Scraping, Pandas* — **Dec. 2024 – Jan. 2025**

- Built a 2013–2025 data pipeline to predict EPL standings using automated data collection and machine learning.
- Programmed custom Python scripts to scrape, clean, and aggregate season-by-season data from fbref.com.
- Applied predictive models (Random Forests) for the rest of season teams' standings and performance with automated visualization of results using Plotly and Matplotlib.